East West University
Department of Computer Science and Engineering
Course Outline
Summer 2025 Semester

Course: CSE106 Discrete Mathematics

Credits and Teaching Scheme

	Theory	Laboratory	Total
Credits	3	0	3
Contact Hours	2.5 Hours/Week for 15 Weeks + Final Exam in the 16th Week		2.5 Hours/Week for 15 Weeks + Final Exam in the 16th Week

Prerequisite

None

Instructor Information

Instructor: Md Ashraful Haider Chowdhury, Lecturer, Dept of CSE.
Office: Room #1003 (AB3 Building)
Tel. No.: To be announced later
E-mail: ashraful.chowdhury@ewubd.edu
TA: Md. Lauhe Mahfus Labib (2022-1-60-030@std.ewubd.edu)

Office Hour

Day	Time
Sunday	**01:30 PM – 3:00 PM**
Monday	**10:10 AM – 01:20 PM**
Wednesday	**10:10 AM – 01:20 PM**
Thursday	**04:50 PM – 6:20 PM**

Course Objective

This course builds up the students' ability to think and express logically and mathematically. The course will address mathematical reasoning, combinatorial analysis, algorithmic thinking, and discrete structures. Knowledge of this course will be needed as prerequisite knowledge for future courses such as CSE110 Objected Oriented Programming, CSE207 Data Structures, CSE246 Algorithms, CSE302 Database Systems, CSE366 Artificial Intelligence, and CSE405 Computer Networks.

Knowledge Profile

K2: Conceptually-based mathematics, numerical analysis, statistics, and formal aspects of computer and information science
K3: Theory-based engineering fundamentals

Learning Domains

Cognitive - C2: Understanding, C3: Applying
Psychomotor - P3: Precision
Affective - A2: Responding

Program Outcomes (POs)

PO1: Engineering Knowledge

Complex Engineering Problem Solution

EP1: Depth of knowledge required
EP2: Range of conflicting requirements

Complex Engineering Activities

None

Course Outcomes (COs) with Mappings

After completion of this course students will be able to:

CO	CO Description	PO	Learning Domains	Knowledge Profile	Complex Engineering Problem Solving/ Engineering Activities
CO1	**Interpret** and **apply** propositional logic, predicate logic, and theorem proving for mathematical reasoning.	PO1	C2, C3	K2	-
CO2	**Interpret** and **apply** counting principles for combinatorial analysis.	PO1	C2, C3	K2	-
CO3	**Interpret** and **apply** the growth of functions, complexity analysis of algorithms, and integer algorithms for algorithmic thinking; **demonstrate** this knowledge and **write** report for realistic problem solving.	PO1	C2, C3 P3 A2	K2, K3	EP1, EP2
CO4	**Interpret** and **apply** discrete structures such as sets, functions, relations, graphs, and trees for modeling discrete objects; **demonstrate** this knowledge and **write** report for realistic problem solving.	PO1	C2, C3 P3 A2	K2, K3	EP1, EP2

Course Topics, Teaching-Learning Method, and Assessment Scheme

Course Topic	Teaching-Learning Method	CO	Class Tests			Mid Semester Exam		Presentation	Project	Final Exam	
			1	2	3						
			C2	C2	C2	C2	C3			C2	C3
Propositional Logic, Propositional Equivalences, Predicates and Quantifiers, Nested Quantifiers	Lectures and discussions inside and outside the class	CO1	5			7 [3+4]	8 [4+4]				
Introduction to Proofs, Mathematical Induction	Do	CO1					5 [2+3]				
Sets, Set Operations, Functions, Recursive Functions, Relations and Their Properties	Do	CO4				5 [2+3]	5 [2+3]				
The Basics of Counting, The Pigeonhole Principle	Do	CO2		5						10 [5+5]	5 [5]
Algorithms, The Growth of Functions, Complexity of Algorithms, The Integers and Division, Primes, Greatest Common Divisor, Least Common Multiplier	Do	CO3			5			5	5	6 [6]	11 [6+5]
Graphs, Graph Terminologies and Special Types of Graphs, Representing Graphs, Introduction to Trees	Do	CO4							5	6 [3+3]	7 [3+4]
			15			30		5	10	40	

Presentation

Presentation	Teaching-Learning Method	CO	Mark of Cognitive Learning Level	Mark of Psycho-motor Learning Level	Mark of Affective Learning Level	Presentation Mark
			C3	**P3**	**A2**	
Presentation on a selected topic	Self-study and five-minute individual power point presentation	CO3	4	0.5	0.5	5
Date of presentation will be announced in the class						

Mini Projects

Mini Project	Teaching-Learning Method	CO	EP/ EA	Mark of Cognitive Learning Level	Mark of Psycho-motor Learning Level	Mark of Affective Learning Level	CO Mark	Mini Project Mark
				C3	**P3**	**A2**		
Mini Project with reports and presentation	Group-based moderately complex programming projects with report writing and presentation	CO3 CO4	EP1, EP2 EP1, EP2	4 4	0.5 0.5	0.5 0.5	5 5	10
		Total		8	1	1	10	
Dates of report submission and presentation will be announced in the class								

Overall Assessment Scheme

Assessment Area	CO				PO Marks
	CO1	**CO2**	**CO3**	**CO4**	**PO1**
Class Test	5	5	5		15
Mid Semester Exam	20			10	30
Presentation					
Final Exam		15	17	13	45
Mini Projects with report and presentation			5	5	10
Total Mark	**25**	**20**	**27**	**28**	**100**

Teaching Materials/Equipment

Textbook:

Kenneth H. Rosen, *Discrete Mathematics and Its Applications with Combinatorics and Graph Theory*, 7th Edition, Tata McGraw-Hill Publishing Company Limited, New Delhi, 2015.

Mini Projects:

Mini Project description will be provided in the class.

Grading System

Marks (%)	Letter Grade	Grade Point
80% and above	A+	4.00
75% to less than 80%	A	3.75
70% to less than 75%	A-	3.50
65% to less than 70%	B+	3.25
60% to less than 65%	B	3.00
55% to less than 60%	B-	2.75
50% to less than 55%	C+	2.50
45% to less than 50%	C	2.25
40% to less than 45%	D	2.00
Less than 40%	F	0.00

Exam Dates

Section	Midterm	Final
12	Will be notified in the class	See EWU calendar

Academic Code of Conduct

Academic Integrity:

Any form of cheating, plagiarism, personification, falsification of a document as well as any other form of dishonest behavior related to obtaining academic gain or the avoidance of evaluative exercises committed by a student is an academic offence under the Academic Code of Conduct and **may lead to severe penalties as decided by the Disciplinary Committee of the university.**

Special Instructions:

- Students are expected to attend all classes and examinations. A student MUST have at least 80% class attendance to sit for the final exam.
- Students will not be allowed to enter into the classroom after 20 minutes of the starting time.
- For plagiarism, the grade will automatically become zero for that exam/assignment.
- Normally there will be **NO make-up exam**. However, in case of **severe illness, death of any family member, any family emergency, or any humanitarian ground**, if a student miss the **mid-semester exam**, the student MUST get approval of makeup exam by written application

to the Chairperson through the Course Instructor **within 48 hours** of the exam time. Proper supporting documents in favor of the reason of missing the exam have to be presented with the application. **It is the responsibility of the student to arrange a Makeup Exam in consultation with the Course Instructor**.

- For **final exam**, there will be NO makeup exam. However, in case of **severe illness, death of any family member, any family emergency, or any humanitarian ground**, if a student miss the final exam, the student MUST get approval of **Incomplete Grade** by written application to the Chairperson through the Course Instructor **within 48 hours** of the final exam time. Proper supporting documents in favor of the reason of missing the final exam have to be presented with the application. **It is the responsibility of the student to arrange an Incomplete Exam within the deadline mentioned in the Academic Calendar in consultation with the Course Instructor**.
- All mobile phones MUST be turned to silent mode during class and exam period.
- There is **zero tolerance for cheating** in exam. Students caught with cheat sheets in their possession, whether used or not; writing on the palm of hand, back of calculators, chairs or nearby walls; copying from cheat sheets or other cheat sources; copying from other examinee, etc. would be treated as cheating in the exam hall. The only penalty for cheating is **expulsion for several semesters as decided by the Disciplinary Committee of the university**.